

# **ECON 100B: Macroeconomics**

**Class 15: Recent and Not-so-recent “Macro-clysms”  
(Jones Chapter 10)**

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March 17, 2026**

# Agenda

- Announcements
- Current events
- Way way way past events: The Great Famine in Ireland
  - See for example Q3 on the Spring 2014 midterm
- Jones Chapter 10: The Great Recession and the Covid-19 Pandemic: A First Look

# Announcements

- Take-home Midterm Essay released 3/8, due 3/20
  - Scores will be high
  - You can use many tools including AI, but essays written solely by AI typically lose points and break the rules. Up to 10% of ECON 157 for example. AI is not a student in this classroom and won't know the best answers
- Spring Break 3/23-3/27

# Current events

- [Last Friday, a federal judge threw out subpoenas](#) the Justice Department had sent to the Federal Reserve
  - “There is abundant evidence that the subpoenas’ dominant (if not sole) purpose is to harass and pressure Powell either to yield to the President or to resign and make way for a Fed Chair who will,”
- The U.S. Attorney pledged to appeal the ruling
- [WSJ Editorial Board](#):

“The chronology of events suggests that U.S. Attorney Jeanine Pirro was responding to pressure from Mr. Trump to target Mr. Powell in a criminal case without evidence of criminal behavior.

“What matters to the economy is monetary policy, not more harassment of a public servant who on all of the available evidence has done nothing illegal.”





Blarney Castle

Cliffs of Moher

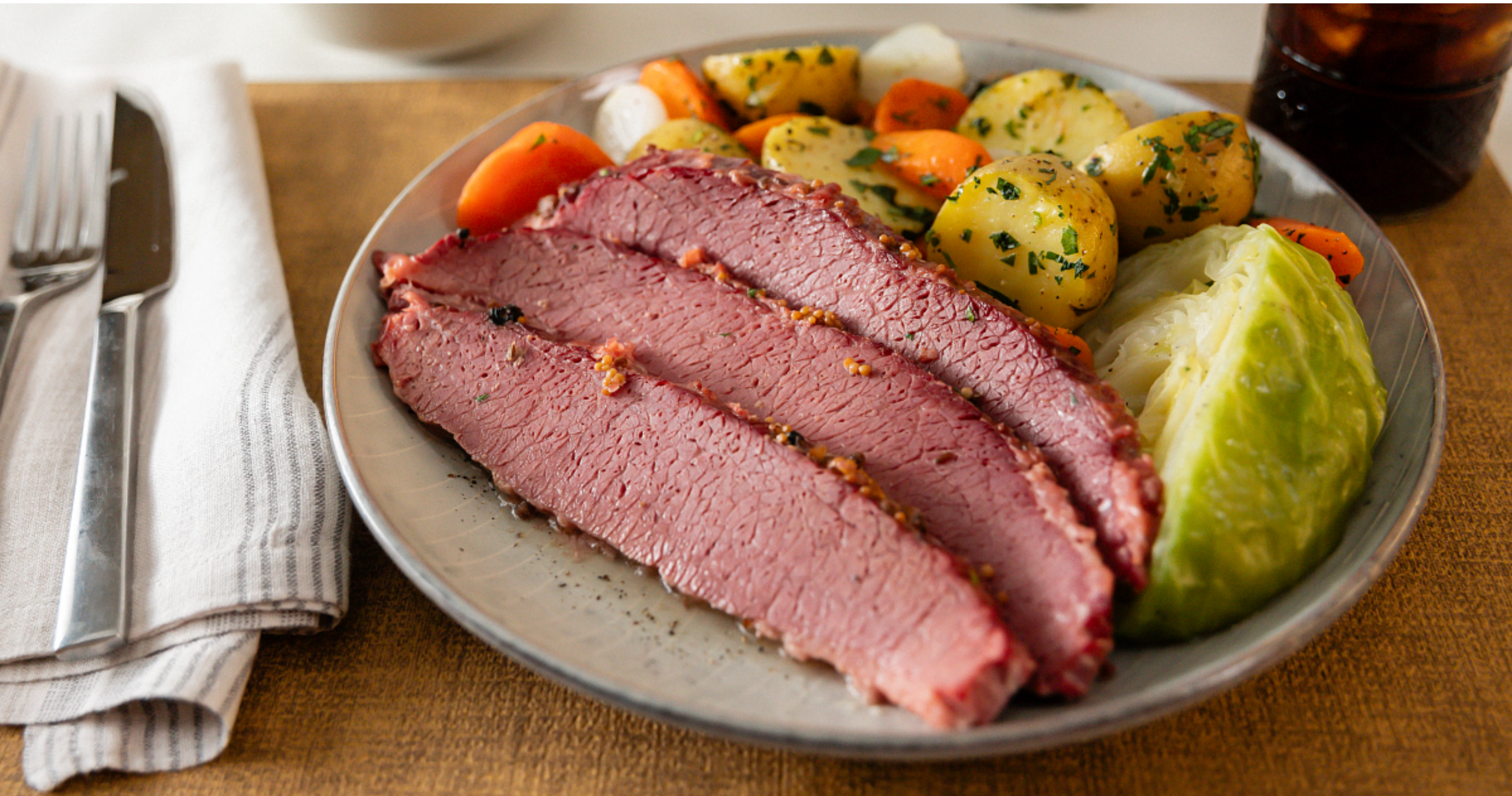


Kissing the  
Blarney Stone





**Pro tip: Don't ask an Irish person what this is**





**“Corned Beef and Cabbage for St. Patrick’s Day? Not So Irish, Historians Say” (NYT 3/17/16)**



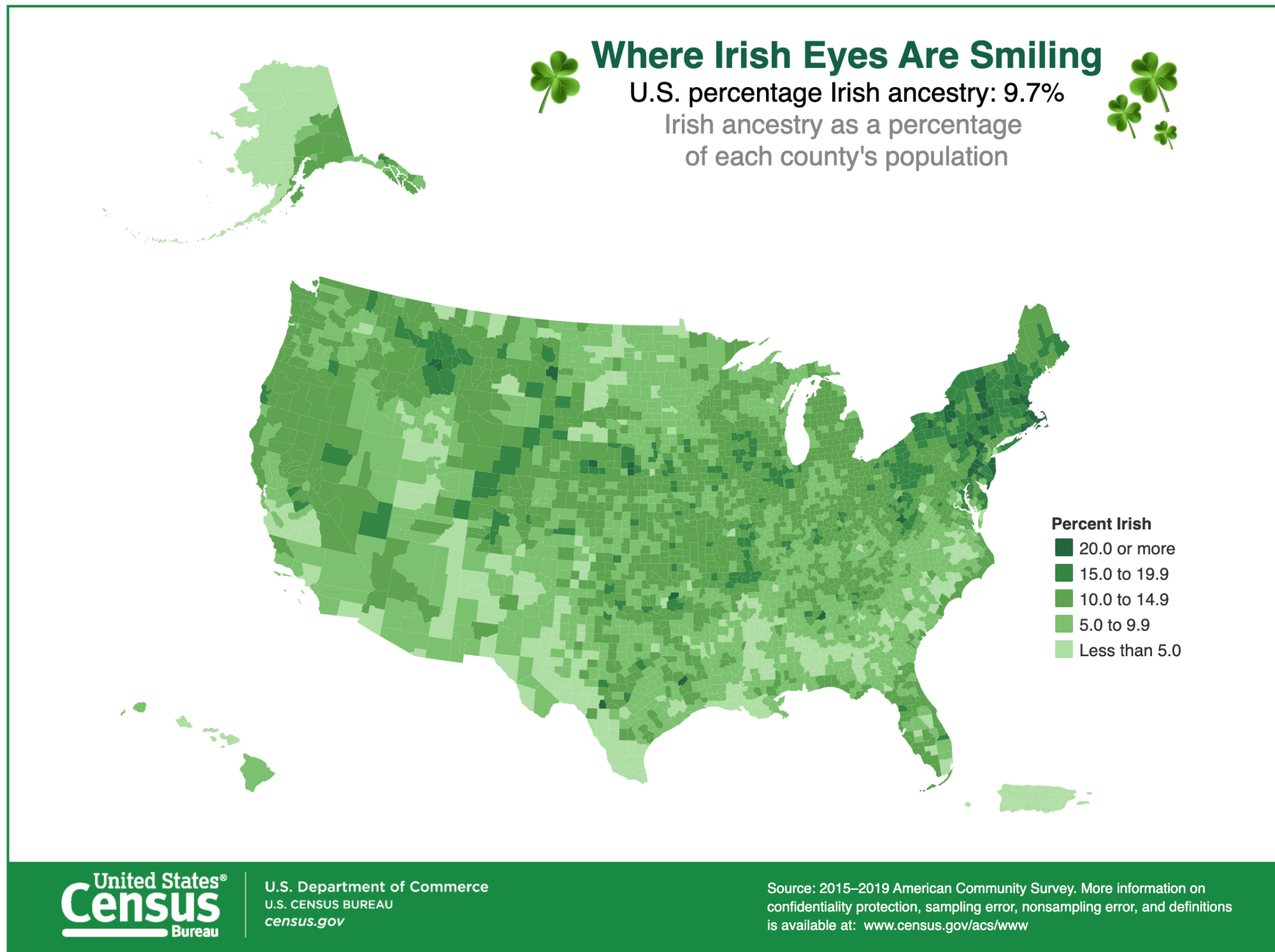
Dennis Dunn stood in what he said was his usual spot on Fifth Avenue at the Saint Patrick's Day Parade in Manhattan in 2015. Ruth Fremson/The New York Times

# Happy St. Patrick's Day!

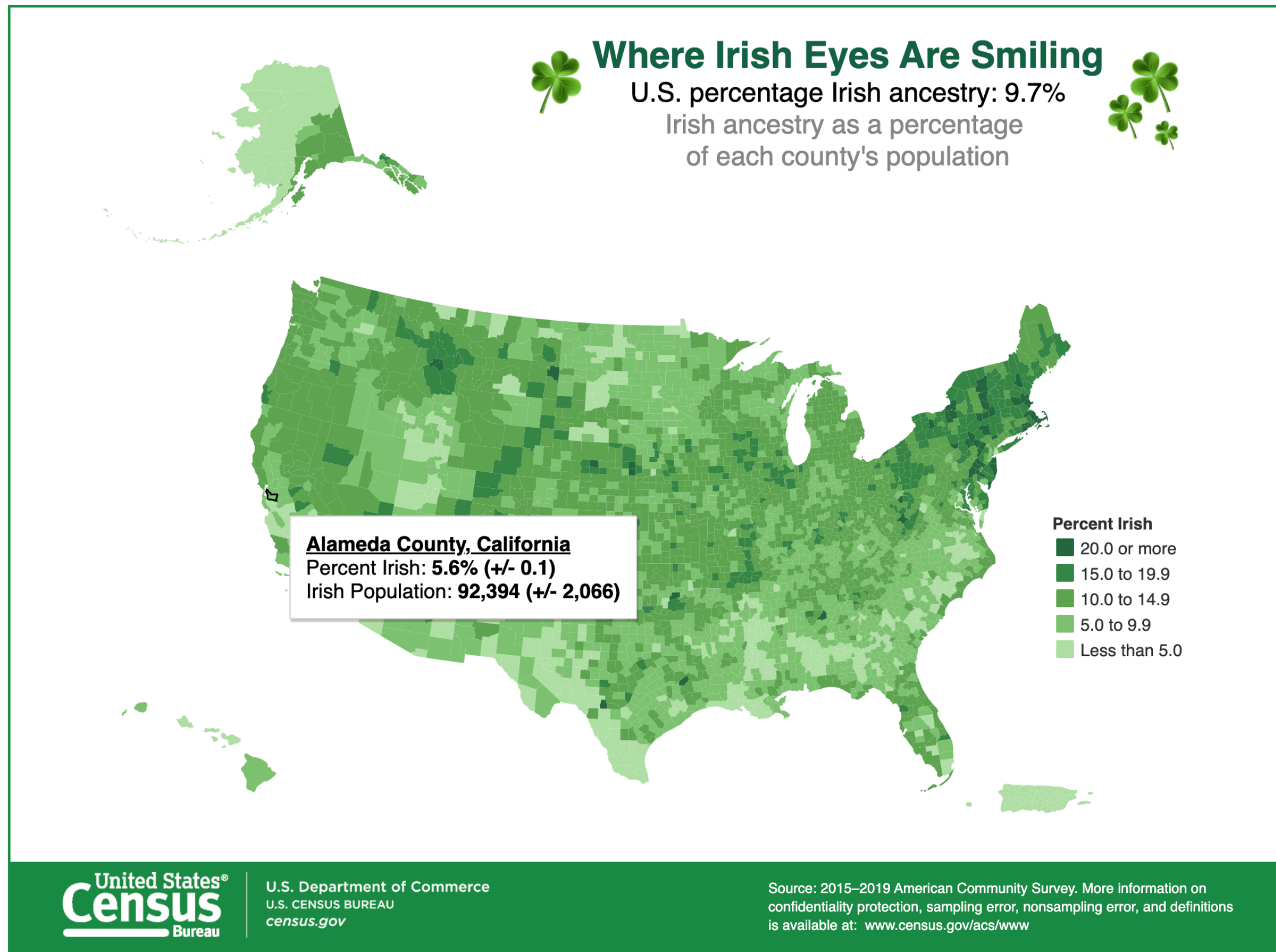
- In 2011, 34.5 million U.S. residents claimed Irish ancestry;  
[in 2019, 31.5 million](#) or about 10%
- This is a little below levels in 1980 and 1990 reported by [Hout and Goldstein \(Am Soc Rev, 1994\)](#), of 40.2 and 38.7 million
- Still, in 2011, there were 7 Irish Americans for every single resident of Ireland, which had a population of 4.6 million
- What events could have led to such a situation?
  - Great Famine of 1846–47 in Ireland caused by a massive failure of the potato crop due to infestation by a potato blight, disease kills potatoes
  - Irish population peaked around 8 million in 1841, fell below 6 million by 1861, reached 4.5 million by 1901
  - Total Irish immigration to the U.S. between 1790 and 1990: 4.8 million per Hout and Goldstein, most arriving between 1820 and 1920



# Census on Irish ancestry in the American Community Survey



# Census on Irish ancestry in the American Community Survey



# Economic demography & Irish history 1

- So today, there are also about **7 times** as many Irish Americans as there ever were actual Irish immigrants to the U.S.
- Hout and Goldstein show that this is more than would be implied by Irish fertility alone (which wasn't particularly high)
  - Irish were equally divided between Protestant and Catholic, Germans too
  - The mix meant Irish and Germans had access to more potential mates of the same religion but from different ethnic groups
  - More intermarriage, and more “subjective ‘closeness’ to Ireland” resulted in more self-identification

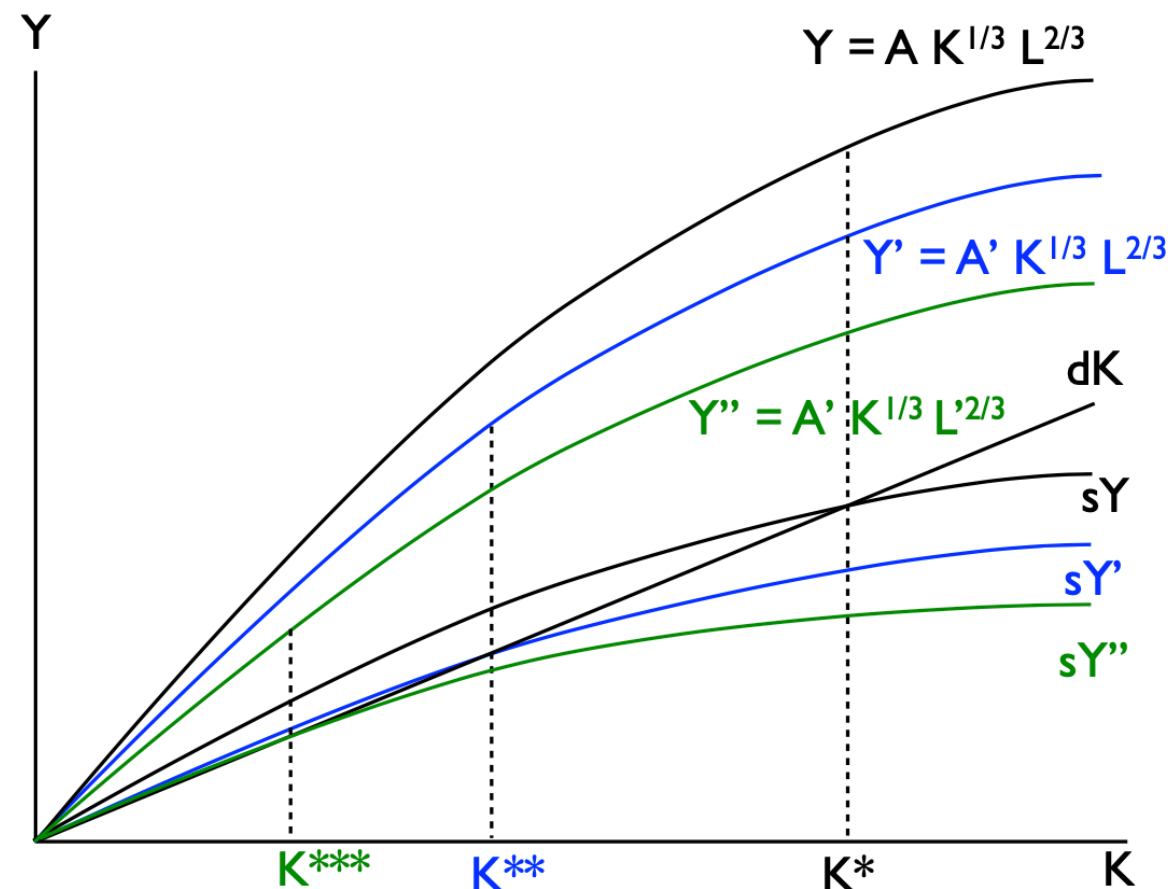
# Economic demography & Irish history 2

- The potato famine event makes for a good exam question
- What happens when the fruits and vegetables of one's labor fall victim to blight?
- What would models like Solow and Malthus tell us about wages?
- Why might Irish immigration to the U.S. accelerate after the blight? The blight doesn't kill people, after all
- What effects might increased Irish immigration have had on the Irish economy? On the U.S. economy?



# Spring 2014 midterm exam

- 3(f) (4 points) Supposing that a large number of Irish residents took the advice you gave in part (e), illustrate the joint effects of that change along with the original change you showed in part (c), and explain. Discuss the net effect on income ( $Y$ ).



**Answer:** If a bunch of Irish workers migrated out, that would reduce  $L$  in the Solow Model. A reduction in  $L$  looks exactly the same as a reduction in  $A$ : they both shift the income and saving curves down simultaneously. The net effect on income is a BIG reduction.

# STAT 20 or ECON 140 case study

- Guinness is to Ireland as Budweiser is to U.S.  
(both are actually now foreign owned)
- [Ziliak \(2008\) “Guinnessometrics:  
The Economic Foundation of Student’s t”](#)
  - William Gosset (1876-1937) was master brewer and experimental scientist with Guinness
  - The problem: With only small samples of malt or hops, how do you infer the quality of the rest, and thus of the beer shipped around the world?
  - His work, published under the pseudonym “Student,” inspired Ronald Fisher to apply the t-statistic to inference



# An aside: Population dependency ratios

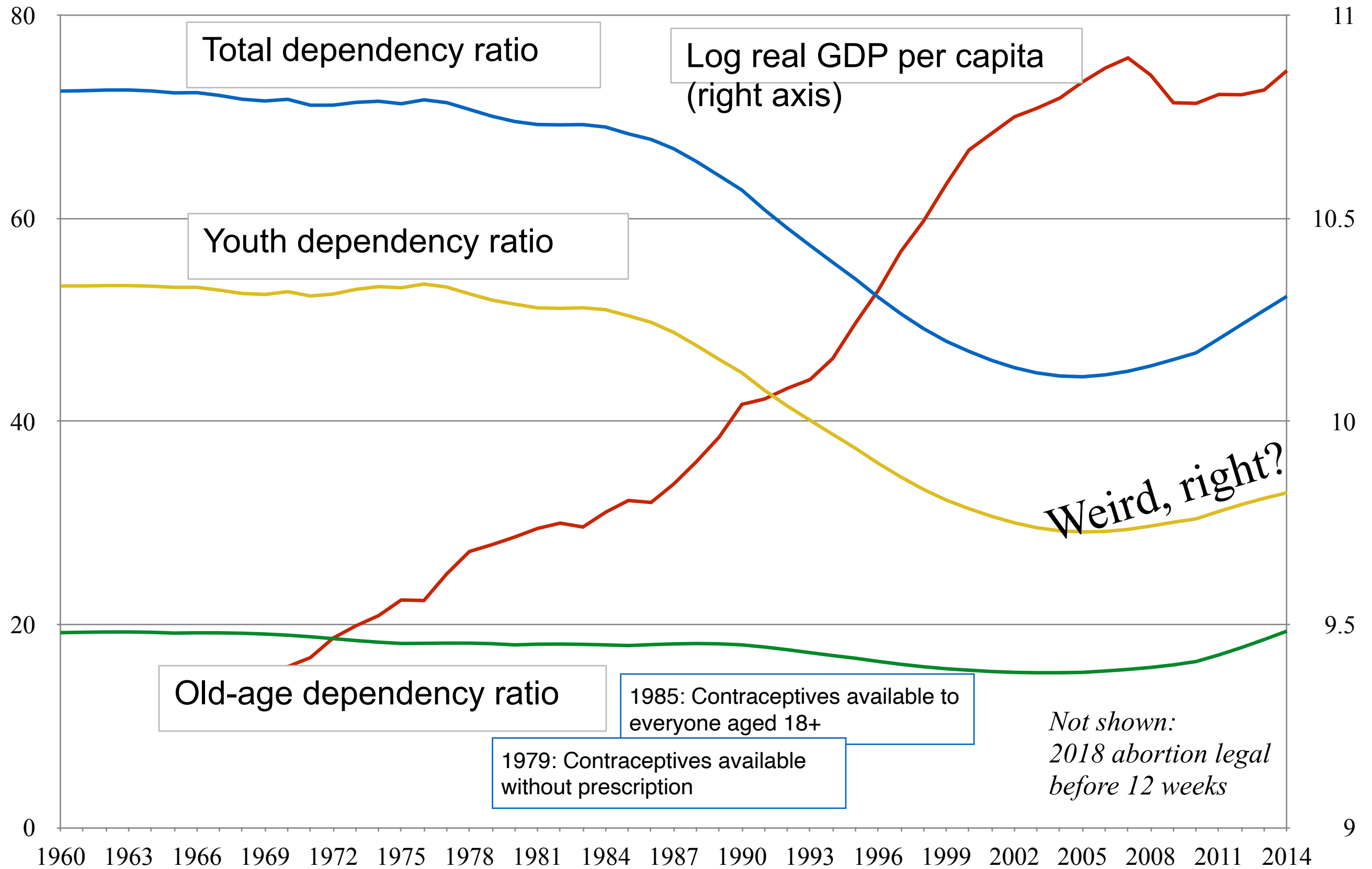
- A different way of saying “age group ratios” where the denominator is the working-age population

$$\text{Youth dependency ratio } YDR = \frac{\text{ages } 0 - 14}{\text{ages } 15 - 64}$$

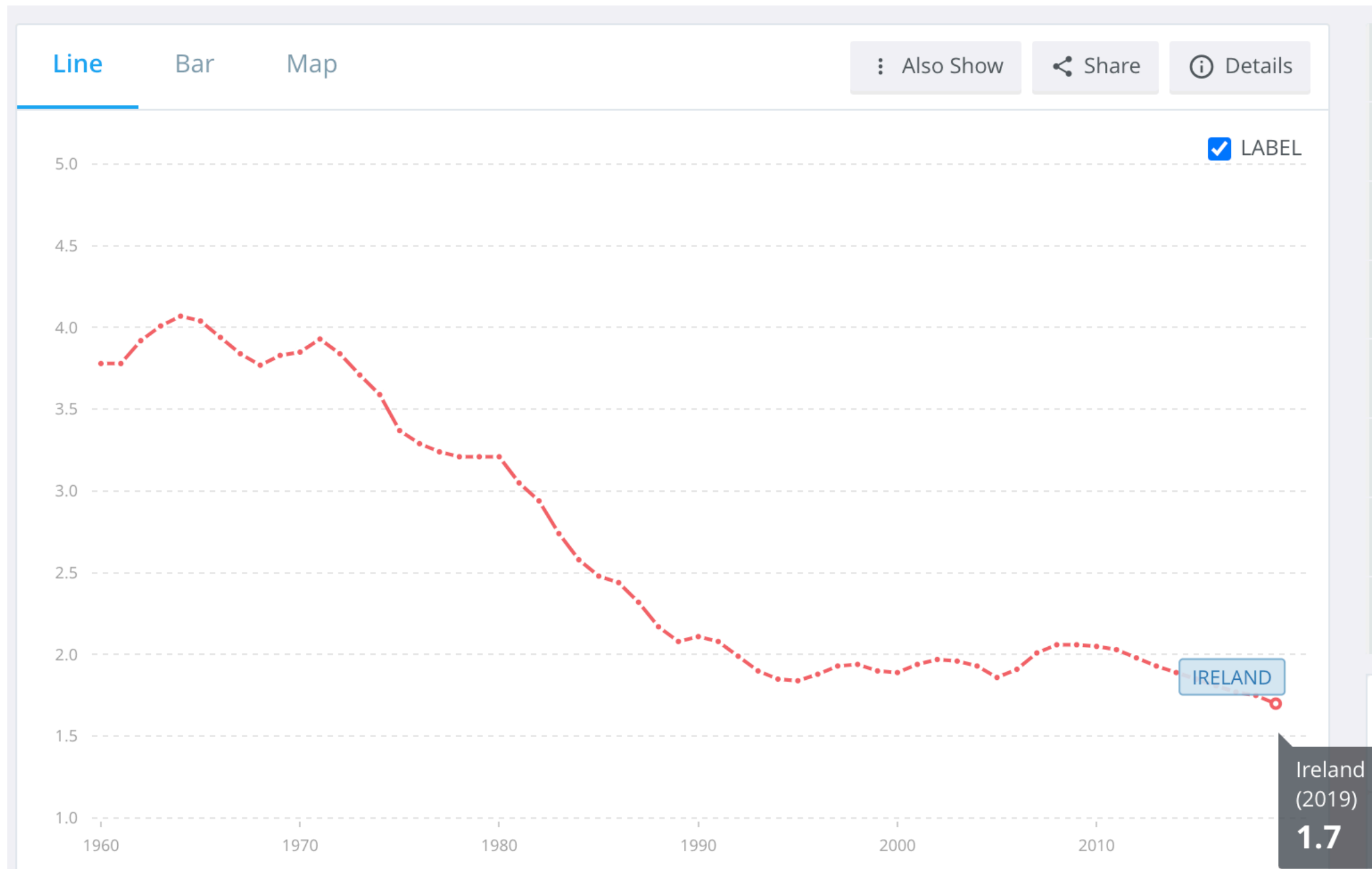
$$\text{Old-age dependency ratio } OADR = \frac{\text{ages } 65+}{\text{ages } 15 - 64}$$



# More Irish history and DEMOG/ECON c175

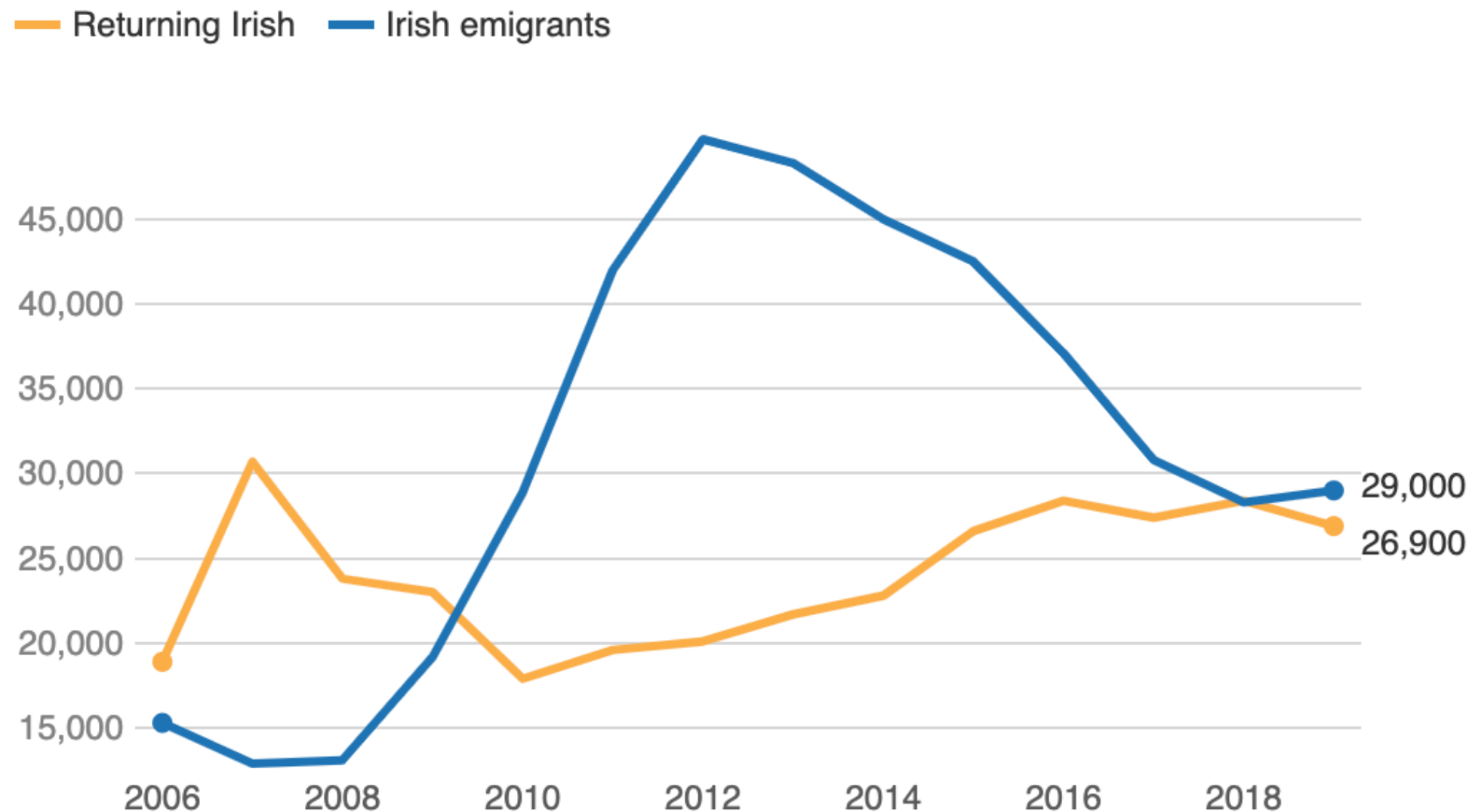


# It's not a rise in the fertility rate



# “More Irish emigrants leaving the country than returning again”

## Irish emigrants and immigrants (returning Irish), 2006-2018



Values are for the 12 months to April each year. Note: these figures are based on data published by the Central Statistics Office in August 2019, including revised figures for 2012-16 based on Census 2016 data.

Source: CSO • Created with Datawrapper



# Where we are and where we're headed

- Chapters 1-8 are about the long run
  - (Supply) Labor, capital, total factor productivity create **potential output**  $\bar{Y}$
  - Social trends and human capital accumulation create a labor force
  - Given  $\bar{Y}$ , money and credit create a long-run rate of inflation,  $\bar{\pi}$ , because  $MV = PY$
- Chapters 9-20 are (mostly) about the short run
  - (Demand) Expenditures can deviate from **potential output**  $\bar{Y}$

$$Y = C + I + G + NX \lesseqgtr \bar{Y}$$

- A big reason is because **investment** responds to the real interest rate. Another is because all components respond to shocks. Those get amplified when consumption responds to current income
- When  $Y > \bar{Y}$ , the economy is booming, and  $\tilde{Y} = \frac{Y - \bar{Y}}{\bar{Y}} > 0$  and the **Phillips Curve** shows us  $\Delta\pi > 0$ . When the economy is slumping: the reverse

# Learning objectives for Chapter 10

- We've seen major macroeconomic shocks recently and in history. Here are some we're discussing; there are many more
  - Black Death in England, 1348-1350, killed perhaps 33%
  - Great Famine in Ireland, 1845-1852
  - Great Depression worldwide, 1929-1933 or so
  - Great Recession worldwide, 2007-2009 or so
  - COVID19 pandemic, 2020
- A big part of ECON 100B is understanding how **monetary** and **fiscal policies** by government can help
- Financial regulation and lender-of-last-resort activities also matter

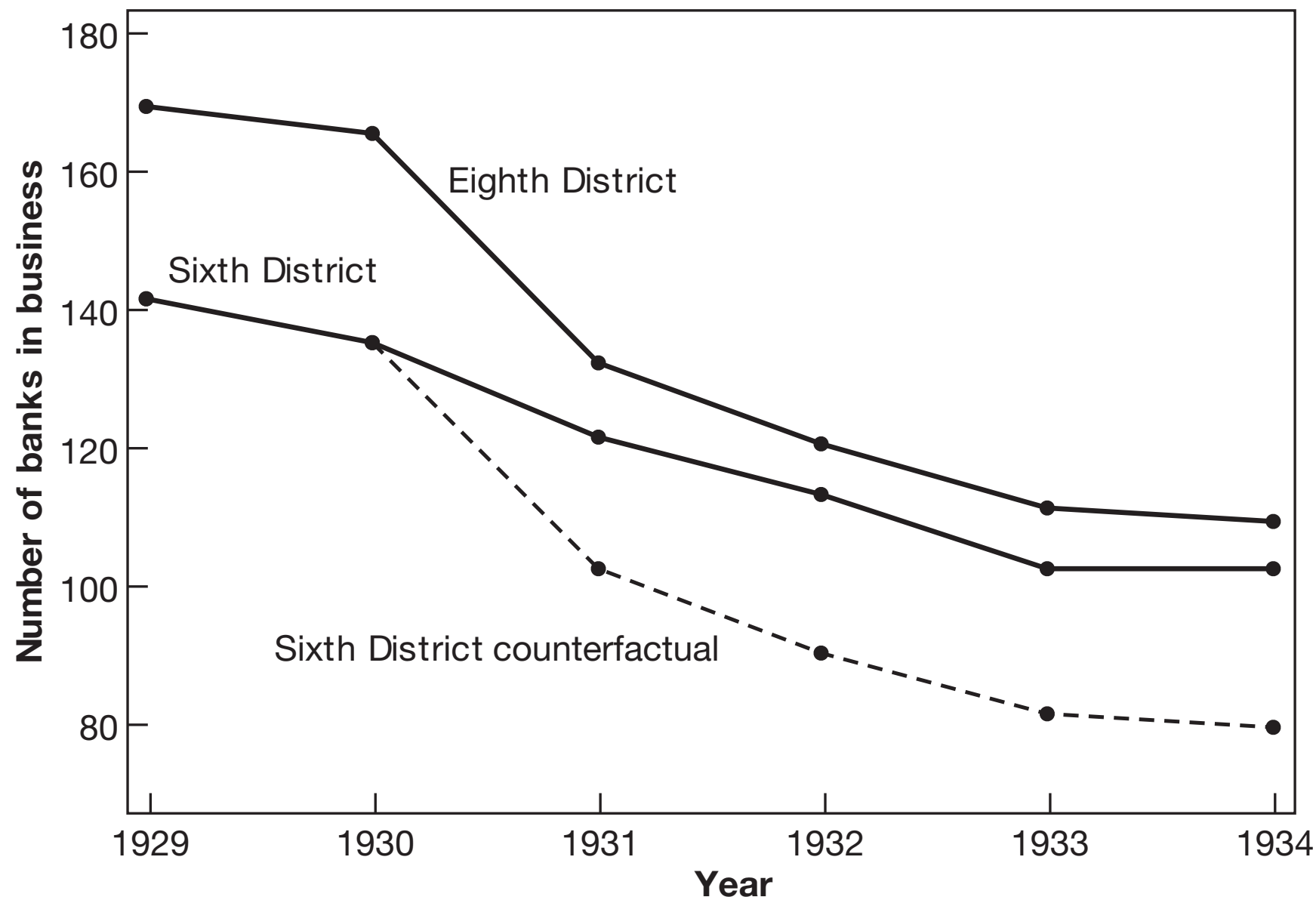
# $MV = PY$ is useful for the long run and historical settings

- Modern central banking employs different tactics
- But especially in historical periods,  $MV = PY$  reveals some important truths
- The money supply  $M$  is more than cash; it's really the amount of credit readily available
  - [Wikipedia on monetary aggregates](#); and a [11/12/24 Fed update](#)
  - $M0$  is what might first think of: coins and currency
  - $M1$  includes checking and saving deposits;  $M2$  is less liquid still, etc.
  - **Banks are important for  $M$**

# ECON 140: Banks collapsed during the Great Depression

FIGURE 5.3

Trends in bank failures in the Sixth and Eighth Federal Reserve Districts, and the Sixth District's DD counterfactual



In the Sixth Fed District, policymakers acted as lenders of last resort

In the Eighth Fed District, policymakers let banks burn

MM makes the point that after 1931, Fed banks coalesced more around bailouts, and that's why we see similar time trends there

*Notes:* This figure adds DD counterfactual outcomes to the banking data plotted in Figure 5.2. The dashed line depicts the counterfactual evolution of the number of banks in the Sixth District if the same number of banks had failed in that district after 1930 as did in the Eighth.



# Bank bailouts

- There are very real costs and benefits of bank bailouts
- Costs: Increased **moral hazard**
  - Banks that know they'll be bailed out are likely to behave with more risk than they normally would. Is that socially optimal?
- Benefits: reduced systemic risk
  - Banks on "Wall Street" create the financial ecosystem that we on "Main Street" use every day
  - Banks may deserve to fail, but if they take us with them, that stinks

**In Ireland's Great Famine (1845-1852) the money supply fell around 30% very quickly. In the U.S. during the Great Depression, it fell like that later**



*Figure 1. The Great Famine versus the Great Contraction*

*Notes: Ireland indexed to 1846 = 100. US indexed to 1928 = 100. Year end.*

*Sources: Kenny and Lennard, 'Monetary aggregates for Ireland'; Friedman and Schwartz, Monetary history.*

# A look back at ECON 2 Class 19:

## Learning objectives for Short-Run Macro Fluctuations

- Real activity responds to the **real interest rate**  $r$ , which equals the nominal interest rate minus the rate of inflation:  $r = i - \pi$
- The real interest rate is the **price of spending today**, the opportunity cost of what you forgo tomorrow by spending today
- The real interest rate can dip negative, and that is the macroeconomy and policymakers trying to price spending today low enough to get people to increase their spending
- In the macroeconomy, my spending is your income, and your spending is my income. If I stop spending, you might have to
- Can we buffer ourselves against macroeconomic shocks? How much can we draw on our own savings, or our families' savings?

*(Narrator voice: Not much. At least, for most of us.)*

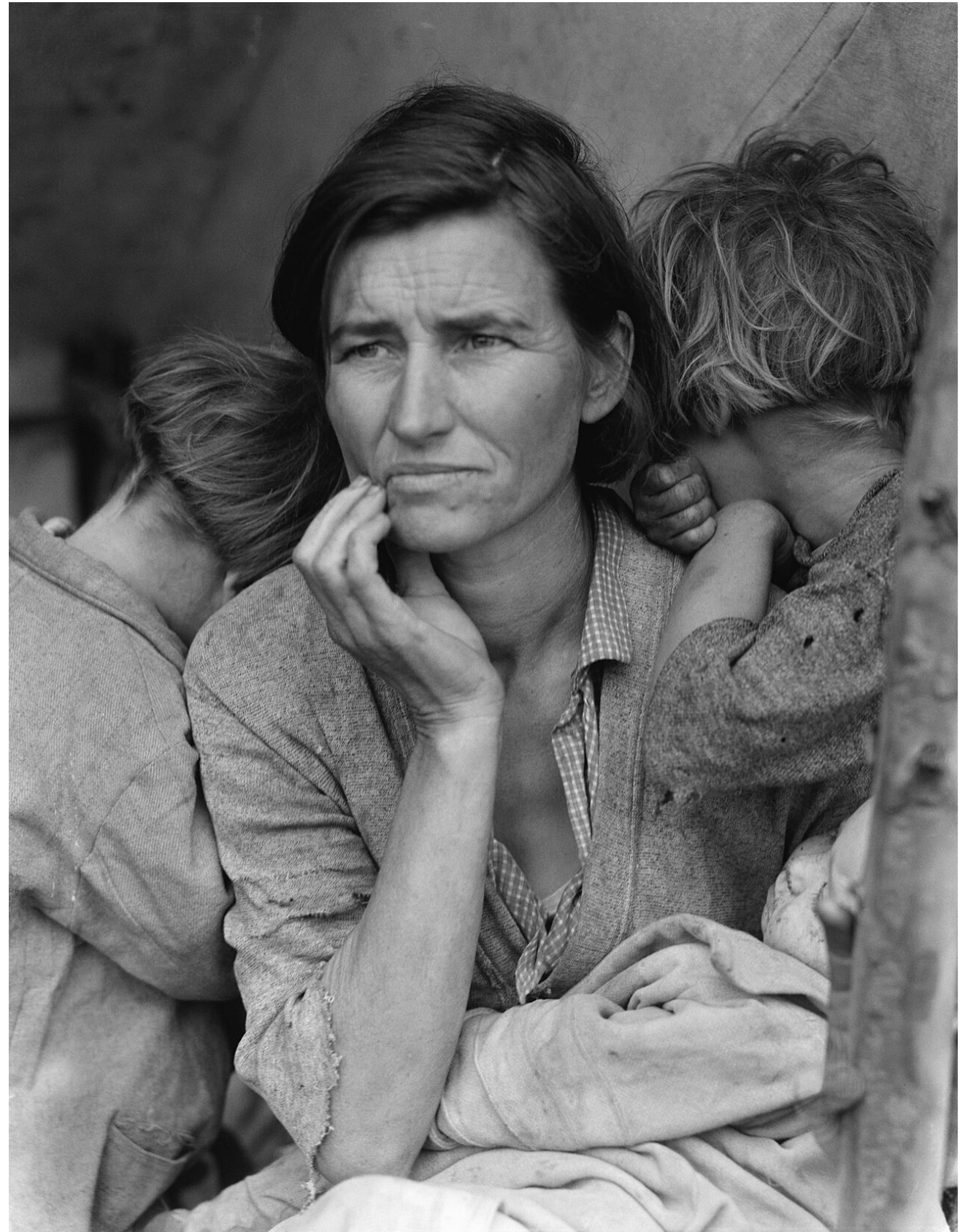
- The **Keynesian Cross** shows us how macroeconomic spending shocks reverberate through the economy

# Why do we care? What's special or unexpected?

- “[Migrant Mother](#)” by Dorothea Lange, March 1936

Nipomo, CA, between Santa Barbara and San Luis Obispo on 101

- Migrant farm workers looking for employment
- Why a **Great Depression**?  
Wasn't it just a stock market crash? Why did that produce widespread human misery?





# Consumer spending collapsed after the stock market crash of October 1929

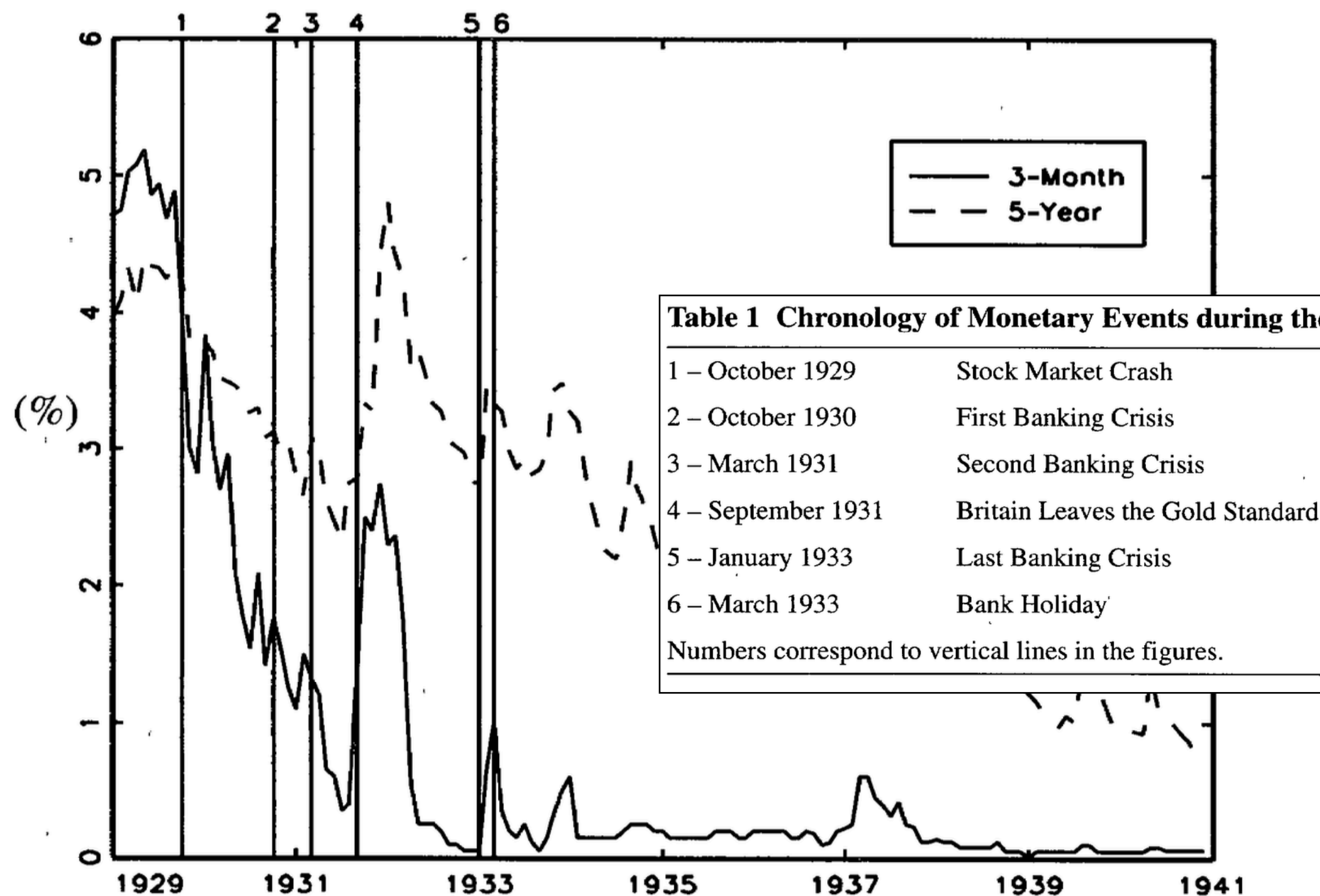
TABLE I  
CONSUMER BEHAVIOR FOLLOWING THE GREAT CRASH

|                          | Cumulative percentage change in real<br>seasonally adjusted retail sales |              |              |              |              |              |
|--------------------------|--------------------------------------------------------------------------|--------------|--------------|--------------|--------------|--------------|
|                          | Oct.<br>1929                                                             | Nov.<br>1929 | Dec.<br>1929 | Jan.<br>1930 | Feb.<br>1930 | Mar.<br>1930 |
| Automobile registrations | −5.5                                                                     | −14.1        | −18.9        | −23.7        | −11.7        | −20.4        |
| Department store sales   | −8.4                                                                     | −10.1        | −4.5         | −15.8        | −11.7        | −16.4        |
| Mail-order sales         | −4.1                                                                     | −7.4         | 3.4          | −20.6        | −25.6        | −35.8        |
| Ten-cent store sales     | −0.3                                                                     | 1.7          | −2.5         | −2.7         | −0.1         | −7.4         |
| Grocery store sales      | 5.9                                                                      | 3.1          | 3.4          | NA           | NA           | NA           |

Source: Christina Romer, “The Great Crash and the Onset of the Great Depression.”

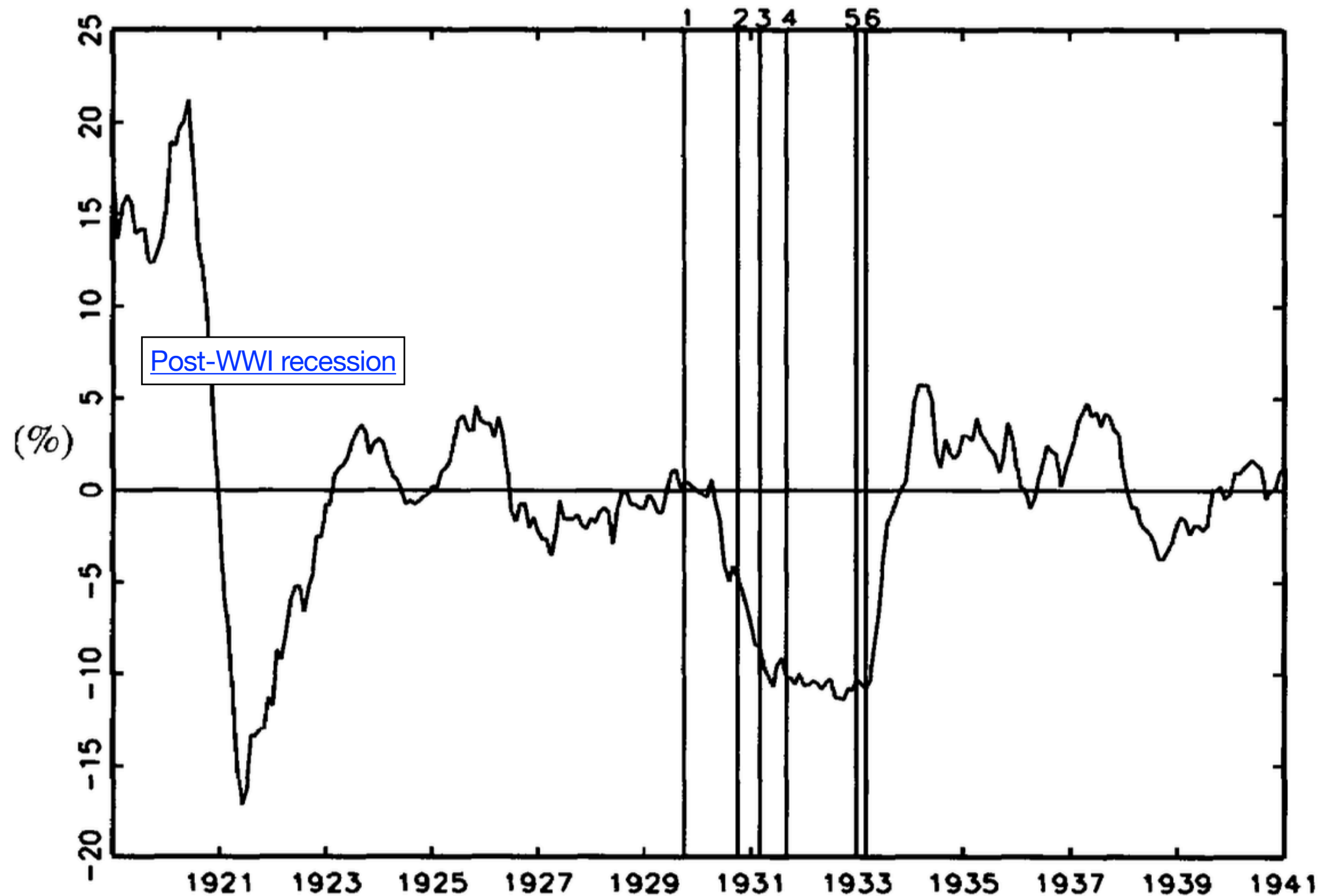
# Nominal interest rates fell ...

**Figure 1 Monthly Nominal Interest Rates, 3-Month and 5-Year Yields, 1929 to 1940**



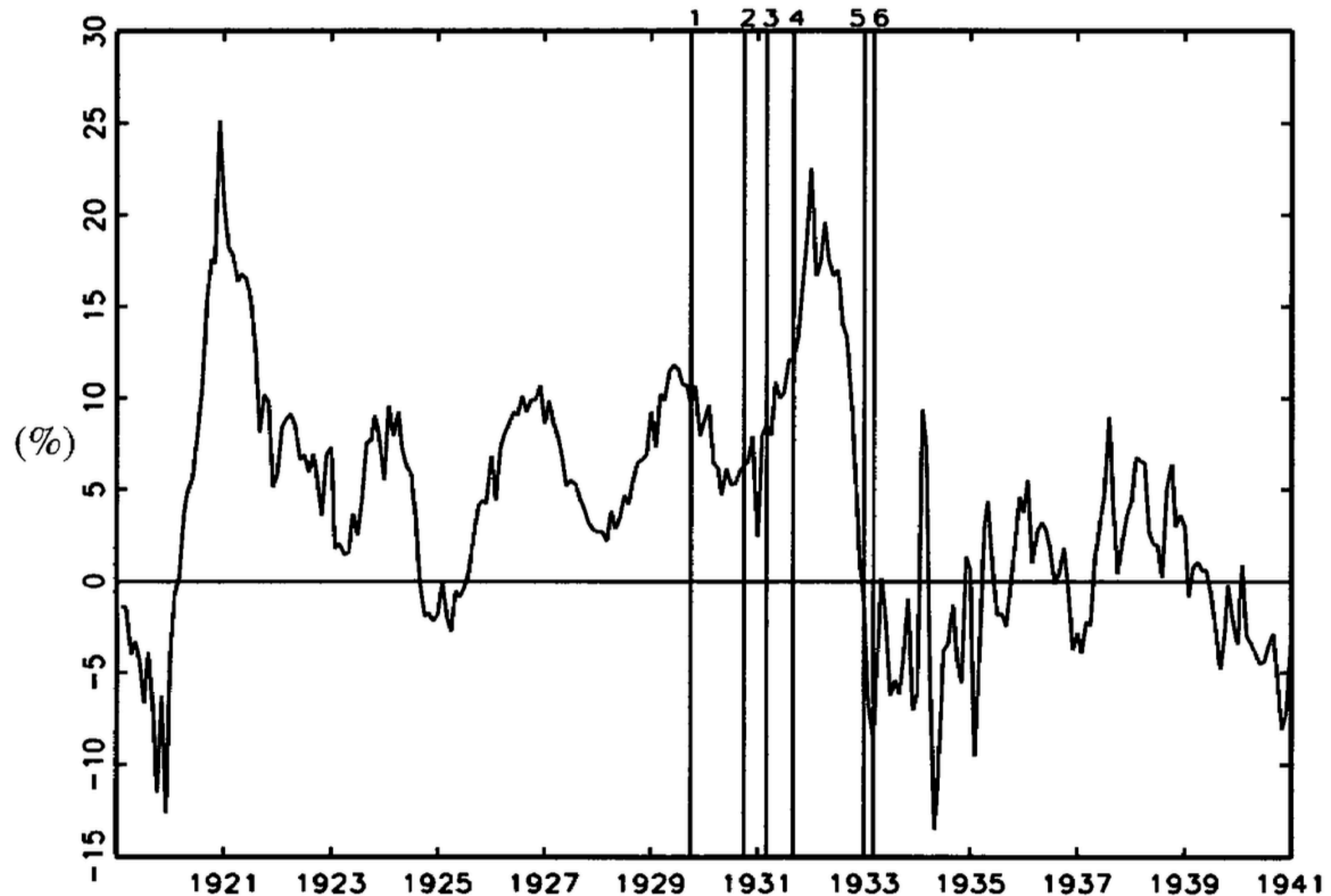
But consumer prices collapsed, and inflation  $\pi$  plummeted

**Figure 3 Monthly Changes in Consumer Price Inflation, 1919 to 1940**



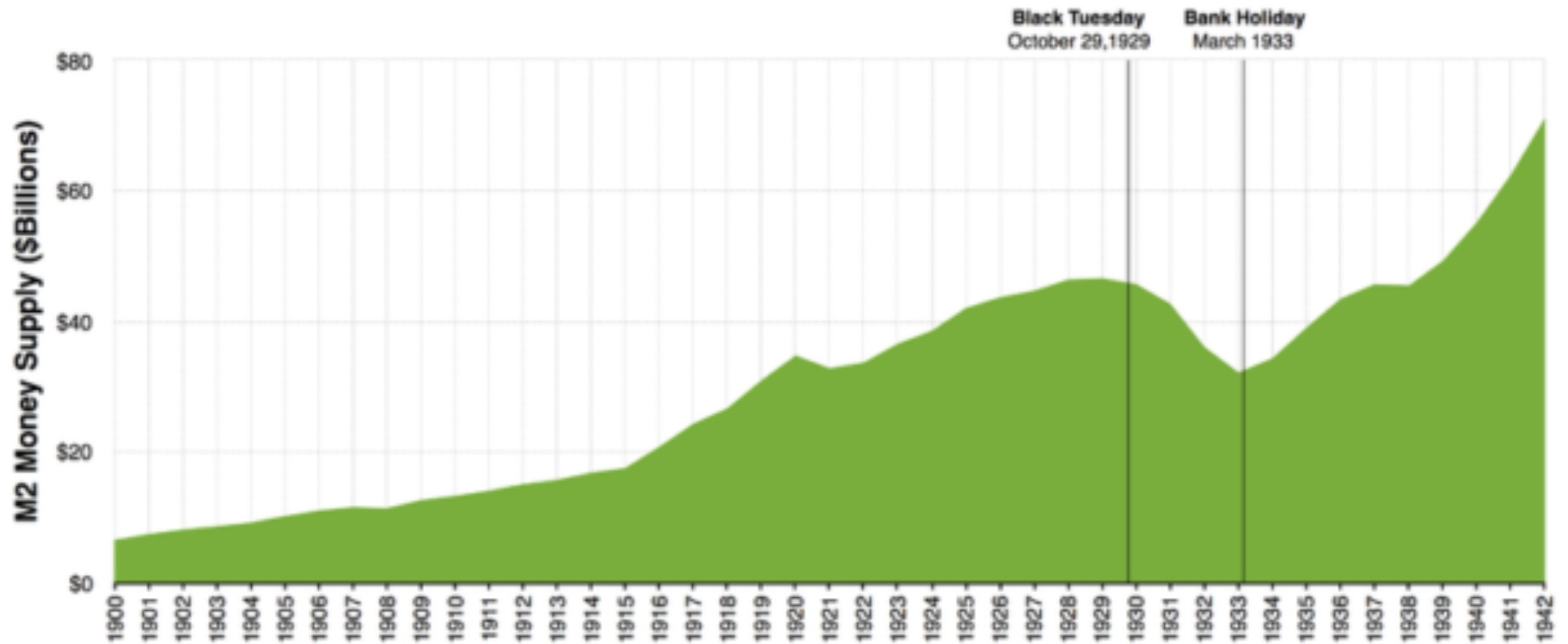
Thus the real interest rate soared higher at first, and that made things much worse

**Figure 4 *Ex ante* Real Interest Rate, 1919 to 1940**  
(monthly at an annual rate, 3-month horizon)





# Money Supply During The Great Depression Era



Source: Dave Henderson, [“The Factors in the Drastic Money Supply Drop from 1929 to 1933”](#)